Copilot in Excel: DMAIC and improvement reporting

90 minutes

IES | Original synthetic CW-401 engineering case. Values and limits are fictional teaching assumptions, not plant operating instructions. Generated references are independently calculated examples, not recorded Copilot executions.

Preparation

Save a working copy of Starter.xlsx to your approved OneDrive or SharePoint location. Open it in Excel with your work/school account and use the Copilot side pane. Select the named table before prompting. If Copilot cannot edit, request the formula and insert it manually; record the fallback. Confirm tenant entitlement and supported workbook format before class. Keep the local raw files unchanged.

1. Define the problem

Open WeeklyEvents and DefectCounts. These are an eight-week event register, separate from the120-test engineering investigation sample. Define the CTQ as acceptance deviations per logged test opportunity. It is not treatment efficacy or regulatory compliance.

2. Calculate exposure-aware rates

Submit DMAIC1. Sum baseline weeks1–6:93 deviations/1008 opportunities=9.22619%. Pilot weeks7–8:11/336=3.27381%. Label phase and denominator. A comparison with unequal observation periods should use rates, not raw count reduction.

3. Create the Pareto

Submit DMAIC2. Counts55,16,12,10 sum to93. Cumulative shares59.14%,76.34%,89.25%,100%. Use descending columns with a cumulative-percent line on a secondary axis. Categories prioritise investigation and do not establish root causes.

4. Interpret change honestly

Absolute reduction=5.95238 percentage points; relative reduction=64.51613%. The pilot has only two weeks, and strainer replacement plus verification occurred together. Do not call the result proof of one cause, permanent improvement or quantified financial savings.

5. Link Analyse to Improve

Use the SPC evidence and FMEA action list to propose a follow-up study: agreed CTQ, fixed test conditions, documented instrument check, comparable baseline/pilot scope and independent engineering review. Mark proposal status; do not prescribe hazardous operations.

6. Write the Control plan

Submit DMAIC3. Specify the metric, source, review frequency, responsible role, signal rule and escalation route under approved site procedures. Keep baseline limits fixed until a justified re-baseline is reviewed. Retain ACT-401-04 as open pending evidence.

7. Present one workbook summary

Prepare six cells: problem, baseline, pilot, evidence, limitations and proposed next decision. Save the rates, Pareto, DMAIC and ControlPlan sheets. Verify every management claim back to a source row and give a two-minute review to a partner.

Copilot prompts

DMAIC1 Define and measure

Use WeeklyEvents, DefectCounts and Evidence. Draft a DMAIC table in Excel: phase, claim, metric, source, limitation, next action. Define the CTQ as delivery-test acceptance deviation, not chemical compliance. Calculate baseline and pilot rates using summed deviations divided by summed opportunities, not a simple average when denominators differ.

DMAIC2 Analyse

Create a Pareto table and chart from baseline DefectCounts, including cumulative percentage. Reconcile the total with baseline WeeklyEvents. Compare the phase rates and express both percentage-point and relative change. Explain that fewer deviations after two concurrent changes do not isolate the cause or establish sustained improvement.

DMAIC3 Improve and control

Draft a controlled follow-up trial and ControlPlan table: metric, source, frequency, owner, signal rule, review response and evidence. Link to SPC signals and FMEA actions. Preserve the open removed-strainer review. Do not claim causal proof, savings or operational approval not supported by the source. End with a six-cell management summary and unresolved questions.